

A Systematic Review of Blockchain-Based Security Architectures for Trusted Cloud Data Sharing: Threat Models and Open Challenges

Anubhab Das¹, Sumit Kumar Banerjee^{2,*}, Soumyajit Dutta³,
Shraoshi Basak⁴

Department of BSc Cyber Security^{1,3,4}

Department of Computer Science & Engineering^{2,*}

Guru Nanak Institute of Technology

Kolkata, WB, India

anubhab7105@gmail.com, sumit.banerjee@gnit.ac.in,
soumyajitofficial98@gmail.com, shraoshibasak.9090@gmail.com

Abstract

Cloud computing has significantly improved data storage and collaborative sharing, but centralized architectures introduce critical vulnerabilities like unauthorized access, data tampering, and single points of failure. These risks become severe in sensitive domains such as healthcare and IoT. Blockchain technology presents a promising decentralized paradigm to enhance the security, transparency, and integrity of cloud-based data sharing without relying on centralized authorities. This paper presents a comprehensive review of blockchain-enabled secure cloud data sharing frameworks published between 2023 and 2025. It systematically analyze existing architectures, including hybrid models, smart contract-based access control, decentralized identity management, and privacy-preserving encryption. Furthermore, the paper investigates common threat models affecting these ecosystems, such as insider threats, Sybil attacks, and smart contract vulnerabilities. Recent solutions are comparatively evaluated based on scalability, security guarantees, and computational overhead. Finally, we identify critical open research challenges—including cross-platform interoperability, blockchain scalability limitations, and integration with emerging technologies like edge computing and federated learning. This review aims to guide future research toward robust, scalable, and trustworthy blockchain-cloud architectures for next-generation distributed environments.

Keywords: Blockchain, Secure Cloud Data Sharing, Access Control, Privacy-Preserving Systems, Threat Modeling, Cloud Security